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SUPERVISED LEARNING



What Kind of Data Types?

Machine Translation



The dataset is a collection of tuples:

$\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(n)}, y^{(n)})\}$ Where:

- ▶ $x^{(i)} = [x_1^{(i)}, x_2^{(i)}, \dots, x_m^{(i)}]$ is a sentence or a sequence of words in the source language.
- ▶ $y =$ the translation of the sentence x in the target language

Example: Translating public health advisories or medical guidelines into different languages.



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What Kind of Data Types?



- **INFORMATION EXTRACTION**

Example—Extracting information about disease outbreaks, medication dosages, or patient symptoms from electronic health records or news articles.

- **NAMED ENTITY RECOGNITION (NER)**

Example—Identifying names of diseases, medications, or public health organizations in a text.

- **TEXT CLASSIFICATION**

Example—Classifying social media posts or news articles into categories like 'disease outbreak', 'vaccine information', 'healthcare policy', etc.

- **SENTIMENT ANALYSIS**

Example—Analyzing public sentiment towards a new health policy or a newly released vaccine.

- **DOCUMENT SUMMARIZATION**

Example—Creating summaries of lengthy health reports or medical research papers.

- **QUESTION ANSWERING**

Example—Answering questions about a medical guideline or a health advisory.